

JIHYEON SEONG

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EDUCATION

Korea Advanced Institute of Science and Technology (KAIST), Republic of Korea

- Ph.D. in Artificial Intelligence, Kim Jaechul Graduate School of AI Sep 2023 – Present
Advisor: Prof. Jaesik Choi · Expected: Aug 2028 or Feb 2029
- M.S. in Artificial Intelligence, Kim Jaechul Graduate School of AI Sep 2021 – Sep 2023
Advisor: Prof. Jaesik Choi

Ewha Womans University, Seoul, Republic of Korea

- B.S. in Computer Engineering Mar 2016 – Feb 2021
- CS Global Field Training, summer exchange student at EPITA, Paris, France Jul 2017

RESEARCH INTERESTS

- Uncertainty quantification under distribution shift [C4]
- Time-series representation learning under change points [C2, C3]
- Explainable AI for clinical decision support [J1, J2, C1]

PUBLICATIONS

C: conference · J: journal · *: equal contribution (co-first author)

- [C4] DistMatch: Adaptive Binning via Distribution Matching for Robust Sequential Conformal Prediction
Enver Menadjeiev*, **Jihyeon Seong***, Jisu Yeo, Jaesik Choi
International Conference on Machine Learning (ICML), 2026 · [openreview:SxBuTatzGe](#)
- [J2] Towards Transparent Time Series Analysis: Exploring Methods and Enhancing Interpretability
Youngjin Park, Anh Tong, Sehyun Lee, **Jihyeon Seong**, Qin Xie, Jaesik Choi
ACM Computing Surveys, 2026 · [doi:10.1145/3794839](#)
Role: wrote “Limitation of XAI in Time Series”
- [J1] Validation of an Acute Kidney Injury Prediction Model as a Clinical Decision Support System
Giae Yun*, Jinyeong Yi*, Sangyub Han, **Jihyeon Seong**, Enver Menadjeiev, Hyunkyung Han, Jaesik Choi, Ji Hyun Kim, Sejoong Kim
Kidney Research and Clinical Practice, 2025 · [doi:10.23876/j.krcp.24.163](#)
Role: model development, software, data curation, methodology
- [C3] Towards Dynamic Trend Filtering through Trend Point Detection with Reinforcement Learning
Jihyeon Seong, Sekwang Oh, Jaesik Choi
International Joint Conference on Artificial Intelligence (IJCAI), 2024, pp. 2324–2332
[doi:10.24963/ijcai.2024/257](#) · [code](#)
- [C2] Towards Diverse Perspective Learning with Selection over Multiple Temporal Poolings
Jihyeon Seong*, Jungmin Kim*, Jaesik Choi
AAAI Conference on Artificial Intelligence (AAAI), 2024 · [doi:10.1609/aaai.v38i8.28743](#) · [code](#)
- [C1] CardioCaps: Attention-Based Capsule Network for Class-Imbalanced Echocardiogram Classification
Hyunkyung Han*, **Jihyeon Seong***, Jaesik Choi
IEEE International Conference on Big Data and Smart Computing (BigComp), 2024, pp. 287–294
[doi:10.1109/BigComp60711.2024.00052](#) · [code](#)

RESEARCH COLLABORATIONS

- Prediction of Acute Kidney Injury** — Seoul National University Bundang Hospital *Apr 2023 – Present*
- Built the residual-CNN prediction model and its layer-wise relevance propagation (LRP) explanations; in a prospective reader study with 11 clinicians, model assistance raised recall from 61.0% to 74.0% and cut mean review time by 8.4 seconds [J1].
- Anomaly Detection in Semiconductor Fabrication** — Samsung Device Solutions *Sep 2022 – Dec 2022*
- Detected and explained contamination events in a semiconductor fab from multivariate time-series sensor data.
- Correction of Electronic Pen Signals** — HiDeep *May 2022 – Aug 2022*
- Designed an autoencoder-based method to correct distortion in electronic pen input signals.
- Deep Air: Blast Furnace Permeability Prediction** — POSCO Gwangyang Works *May 2021 – Dec 2021*
- Designed an explainable CNN-based blast furnace permeability prediction model from operational sensor time series.

HONORS AND AWARDS

- IJCAI Student Travel Grant**, International Joint Conference on Artificial Intelligence 2024
- Best Paper Award**, Korea Computer Congress (KCC) 2023
- “Capsule Neural Networks as Noise Stabilizer for Time Series Data”*
- Digital Idea Market Award** (4th place), Kookmin Bank — ESG AI Investment · code 2022
- Merit Scholarship for Academic Excellence** (ranked 1st in class), Ewha Womans University 2020

ACADEMIC SERVICE

- Teaching Assistant**, Kim Jaechul Graduate School of AI, KAIST
- Interpretability and Interactivity in AI (Prof. Jaesik Choi) *Fall 2025*
 - Deep Learning (Prof. Jaegul Choo) *Spring 2025*
 - Deep Learning (Prof. Jaesik Choi) *Spring 2024*

REFERENCES

Prof. Jaesik Choi — KAIST Kim Jaechul Graduate School of AI, Ph.D. advisor · jaesik.choi@kaist.ac.kr

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